

Linux OS

Unit1

Q1. What is the Linux? Explain main features of Linux?

- **Ans-** Linux is a Unix like Open source Operating System based on Linux kernel.
- It was initially created by Linus Torvalds in 1991.
- It is one of the most popular operating system used in the world due to its speed, security, light weight, free availability.
- Linux is of the reliable operating system for the computer.
- Linux is used in smart phones, Smart T.V., Gaming console.

Main Features of Linux: -

1. **Open Source:** The main advantage of Linux, is it is an open-source operating system. This means the source code is easily available for everyone and any user can use Linux operating system for free without license.

2. **Free:** Another important features of Linux is that it is free to use. This makes it available to everyone.

3. **Security:** Linux is well-known for its security features. Linux is free from various Virus attacks like Spyware, Trojan, Adware etc. Linux is more secure than any other operating system. Linux provides security features like Authentication, authorization, encryption etc. LINUX does not require any anti-virus software.

4. Light weight infrastructure:- Linux is a lightweight operating system that does not require a lot of system resources to run. It requires only 4GB to 8GB of hard disk space to install. It also requires less RAM to run applications. It supports all types of fileformats like text files, audio files, video files, graphic formats etc.

5. Multi-User & Multiprogramming OS : Linux is a multiuser system means multiple users can access system resources like memory, ram, application programs at same time. It is also a multiprogramming system means multiple applications can run at same time.

6.Stability :- It provides high stability. It rarely slows down or freezes and there is no need to reboot it after a short time.

7. Graphical User Interface : LINUX is command-line interface OS. But it also provides interactive graphical user interface like windows. LINUX can be converted into a Graphical User Interface by installing the GUI package. The Popular GUI Package for LINUX is Ubuntu.

9.Portable :- Linux is designed to be highly portable and can run on a wide range of hardware, including desktop computers, servers, supercomputers, and embedded devices.

10. Software Update :- The software updates in Linux are easy and frequent.

11. Compatibility :- Linux is highly compatible with a wide range of hardware and applications. This implies that users may run Linux on nearly any device.

Q2. Difference between GUI and CUI ?

GUI	CUI
(1) GUI stands for Graphical user interface.	(1) CUI stands for Character user interface.
(2) GUI is very friendly and easy to use.	(2) CUI is very confusing and difficult to use.
(3) In GUI more than one task can run at same time.	(3) In CUI only one task can run at a time.
(4) It required more memory.	(4) It required less memory.
(5) Minimum use of keyword.	(5) Maximum use of keyword.
(6) Ex-Windows, Linux	(6) Ex-MS-DOS

Q3. Blocks in Linux OS?

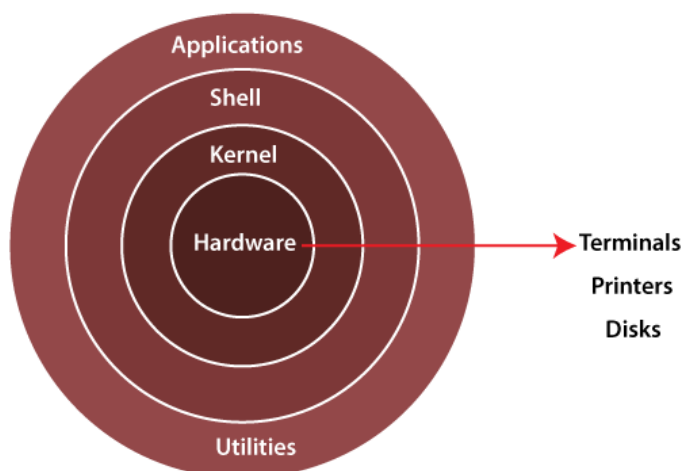
Ans: - In Linux filesystems, blocks are fundamental units of storage used for various purposes within the file system. Here are the main types of blocks in Linux:

- 1. Data Blocks:** Data blocks store the actual content of files, including text, images, programs, etc. When files are written to disk, they are divided into smaller units called blocks, and these blocks are then allocated space on the disk to store the file's data. Data blocks are also used to store filesystem metadata, such as inode information, directory structures, file permissions, timestamps, etc. This metadata is crucial for maintaining the filesystem's integrity and managing file operations.

2. **Inode Blocks:** Inode blocks, contain metadata about files and directories. Each file or directory in a Linux filesystem is represented by an inode, which stores information such as file permissions, ownership, timestamps, and pointers to data blocks. Inodes are crucial for the filesystem to manage files and directories efficiently.
3. **Super Blocks:** Superblocks contain essential metadata about the filesystem itself, such as the total size of the filesystem, the number of blocks available, the block size. Superblocks are typically located at fixed positions within the filesystem and are used during filesystem mounting and maintenance operations.
4. **Boot Blocks:** Boot blocks, also known as boot sectors, are the first sectors of a bootable device (e.g., hard disk, solid-state drive). They contain information required by the system's boot loader to initiate the booting process, including partition table information and details about the filesystem structure.

During the installation of an operating system like Linux, the boot loader (e.g., GRUB) is typically installed in the boot block. The boot loader's primary responsibility is to load the operating system kernel into memory and start its execution.

Q4. Explain Linux Architecture?



The Linux OS architecture contains some of the components the kernel, system library, Hardware Layer, System and Shell utility.

1. **Kernel:** The core component of the Linux operating system. It manages system resources, such as CPU, memory, and peripherals, and provides essential services for other parts of the operating system and user applications.

There are some of the important kernel types which are mentioned below:

- Monolithic Kernel
- Micro kernels
- Exo kernels
- Hybrid kernels

2. **System Libraries:** These are libraries of functions that provide essential program routines for handling input/output operations, memory allocation, system calls, etc. Examples include GNU C Library, which provides the standard C library functions.
3. **System Utilities:** These are essential programs that provide various system management functionalities such as file management, process management, system configuration, and administration. Examples include bash (Bourne Again Shell) and utilities like ls, cp, rm, etc.
4. **Shell:** The command-line interface through which users interact with the operating system. It interprets user commands and executes them. Examples include bash, zsh, and fish.

There are a few types of these shells which are categorized as follows:

- Korn shell
- Bourne shell
- C shell
- POSIX shell

5. **Hardware layer:-** Linux operating system contains a hardware layer that consists of several peripheral devices like CPU, HDD, and RAM.

Unit - 2

Q1. Explain various Linux Commands?

1. cmp:

- `cmp` (compare) command needs two filenames as arguments.
- The two files are compared byte by byte, and the location of the first mismatch is echoed to the screen.
- If two files are identical, `cmp` displays no message, but simply returns the prompt.

Example:

```
user@ubuntu:~$ cat abc
This is some text
user@ubuntu:~$ cat xyz
This is another file
user@ubuntu:~$ cmp abc xyz
abc xyz differ: byte 9, line 1
```

2. comm:

- It requires two sorted files, and lists the differing entries in different columns.
- When you run `comm`, it displays a three-columnar output.
- The **first** column contains lines unique to the first file, and the **second** column shows lines unique to the second file. The **third** column displays lines common to both files.

Example:

```

user@ubuntu:~$ cat file1
Ankur
Charul
Ishaan
user@ubuntu:~$ cat file2
Anubhav
Charul
Himanshu
Vishal
user@ubuntu:~$ comm file1 file2
Ankur
        Anubhav
                Charul
        Himanshu
Ishaan
        Vishal

```

3. diff:

The diff command in Linux is used to compare the contents of two files line by line and display the differences between them.

Symbols used in diff are:

```

a : add
c : change
d : delete

```

Example:

```

user@ubuntu:~$ cat file1
Ankur
Charul
Ishaan
user@ubuntu:~$ cat file2
Anubhav
Charul
Himanshu
Vishal
user@ubuntu:~$ diff file1 file2
1c1
< Ankur
---
> Anubhav
3c3,4
< Ishaan
---
> Himanshu
> Vishal

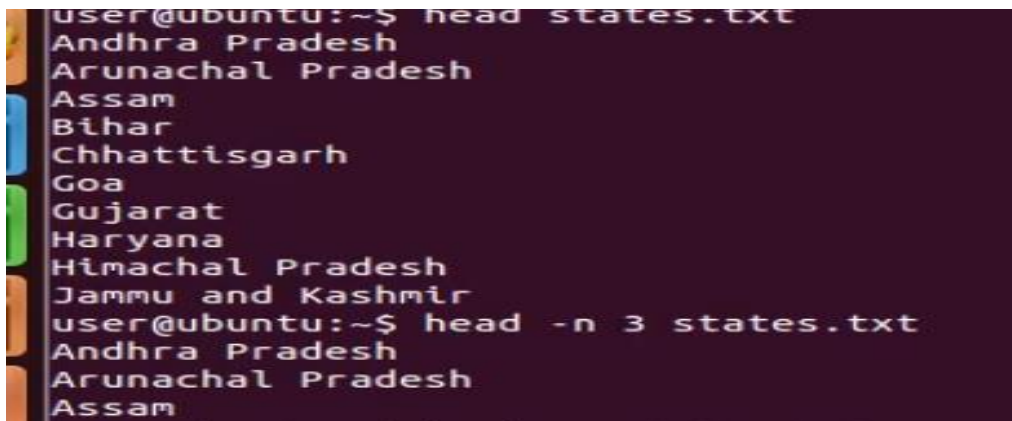
```

4. head:

The head command is used to view the first few lines of a file. By default, it will display the first 10 lines of a file, but this number can be changed with the '-n' option. The syntax for the head command is –

```
$ head [options] [file_name]
```

Example:



```
user@ubuntu:~$ head states.txt
Andhra Pradesh
Arunachal Pradesh
Assam
Bihar
Chhattisgarh
Goa
Gujarat
Haryana
Himachal Pradesh
Jammu and Kashmir
user@ubuntu:~$ head -n 3 states.txt
Andhra Pradesh
Arunachal Pradesh
Assam
```

Several options in head are: -

head -c

- Prints the first 'num' bytes from the file specified.

```
head -c 6 state.txt
Andhra
```

5. tail

The **tail** command is used to display the last few lines of a file. Like the head command, tail will display the last 10 lines of a file by default, but this number can be changed with the **-n** option. The syntax of the tail command is :

```
$ tail [options] [file_name]
```


Example:

```
$ tail -n 3 states.txt
Uttar Pradesh
Uttarakhand
West Bengal
```

Several options in tail are:

tail -c

- Prints the last 'num' bytes from the file specified.

Example:

```
$ tail -c 6 states.txt
Bengal
```

6. cd

cd: Change Directory

The command **cd** uses to change the current working directory to another directory.

cd DirectoryName

Several options in cd are:

Goto home directory

```
cd ~
```

Goto root directory

```
cd /
```

Goto one folder before

```
cd ..
```

cd into a directory with space in the name

```
cd "New Folder"
```

7. cp

cp

Copy files and Directories.

Several options in cp are:

Copy file into directory

```
cp file1 Dir
```

```
cp file1 file2 Dir
```

Copy file content into another file

```
cp file1.txt file2.txt
```

If the file already exists in the directory

```
cp -i file1.txt Dir
```

Copy Directory

```
cp -r dir1 dir2
```

8. mkdir

- mkdir stands for 'make directory'
- mkdir is used to create a new directory.

Syntax: -

```
mkdir dir_name
```

Example:

```
mkdir madan
```

9. rmdir

- rmdir stands for 'remove directory'
- It is used to remove an empty directory.

Example:

```
rmdir madan
```

10 ls

The 'ls' command is used to list files and directories.

simply run the command:

`ls`

There are several options with `ls` command:

i. Display the hidden files and directories

`ls -a`

ii. Display complete information about the files(file permission, size and date)

`ls -l`

iii. To display the files according to his file size

`ls -S`

iv. List all the files and directories in reverse order

`ls -r`

v. To see root directories

`ls /`

11 bc

bc

- `bc` command is used for command line calculator.
- It is similar to basic calculator by using which we can do basic mathematical calculations.

Example:

`bc`

`2+4; 5+10`

`6`

`15`

12. expr

The expr command in linux evaluates a given expression and displays its corresponding output. It is used for:

- Basic operations like addition, subtraction, multiplication, division, and modulus on integers.

```
[root@madan ~]# expr 5 + 5
10
```

13. wc

It used to know the number of lines, characters, words in the file.

First, it shows the number of lines, then no of words, 3rd is the no of characters.

Syntax:

wc file_name

```
[root@madan ~]# wc file10
1 3 15 file10
[root@madan ~]# wc -c file10
15 file10
[root@madan ~]# wc -l file10
1 file10
```

14. more & less command

Both more and less command display the content page by page.

The main difference between more and less command is that in more command we can go to next page but cannot navigate to the previous page.

In less command we can navigate in both direction means go to the next page and navigate to the previous page.

Syntax:

`more file1.txt`

`less file1.txt`

Q2. Explain vi editor?

- The vi editor stands for visual editor.
- The vi editor is a command text editor that is available on almost all Unix systems.
- Vi editor enable us to create and manipulate data on linux file.
- vi is case-sensitive.
- To create a new file using vi just type:

`vi <filename>`

Example:

`vi abc.txt`

vi has different modes for editing text:

1. Command mode: This mode enables you to perform administrative tasks such as saving the files, executing the commands, moving the cursor, cutting and pasting the lines or words, as well as finding and replacing.

To Switch to Command mode press Esc.

2. Insert mode: This mode enables us to insert text into the file. To switch to insert mode press i.

Basic vi commands:

A. Move the cursor

Commands	Action
j	To move down
k	To move up
h	To move left
l	To move right

B. Insert and Replacing text

Command	Description
i	Inserts text before current cursor location
a	Insert text after current cursor location
A	Insert text at the end of current line
o	Creates a new line for text entry below cursor location and switches to insert mode.
O	Creates a new line for text entry above cursor location and switches to insert mode.

C. Copy and Paste the text

yl: copy one character

3yl: copy 3 character

yw: copy one word

5yw: copy 5 words

yy: copy one line

5yy: copy 5 line

D. Searching Text

/string	Forward search for given string
?string	Backward search for given string
/^string	Forward search string at beginning of a line
/string\$	Forward search string at end of a line

E. To save and quit

Commands	Action
:wq	Save and quit
:w	Save
:q	Quit
:w fname	Save as fname
ZZ	Save and quit

F. Deleting text

x: delete one character

5x: delete 5 character

5dw: delete 5 words

p: paste

G. To jump lines

gg: cursor will go to top of the line

5gg: cursor will go to top 5 of line line

G or shiftg: cursor will go to bottom of the file

Q4. Explain File permission in linux ?

In Linux, file permissions refer to the settings that determine who can read, write, and execute files or directories on the system.

Permission Levels in linux:

Each file and directory in Linux has three permission levels. These levels are:

- u - current user
- g - group in which user belong
- o - others
- a - All

Types of permissions:

1. r-read
2. w-write
3. x-execute

Each permission has a numeric value assigned to it:

- r (read): 4
- w (write): 2
- x (execute): 1

In the permission value 744, the first digit corresponds to the user, the second digit to the group, and the third digit to others. By adding up the value of each user classification, you can find the file permissions.

->To Check the Permission of Files in Linux:

ls -l

```
-rw-r--r-- 1 root root 46 Apr 14 16:37 example
```

The 'chmod' command is used to Change Permissions in Linux:

- If we want to add read, write, and execute permissions to the owner, and read and execute permissions to the group and others.

Example: `chmod 755 abc.txt`

- If I want to add read permission to a user then we use '+'.

Example:

`chmod u+r file10`

- If I want to add permission for all.

Example:

`chmod a+rw file50`

or,

`chmod ugo+rw file50`

- If I want to remove read permission to a user we use '-'.

Example:

`chmod u-r file10`

- If I want to remove all permission for all.

Example:

`chmod a-rw file50`

Unit-3

Q1. grep Command in Linux ?

grep stands for “global regular expression print”. Grep command searchfor a particular string/keyword from a file and print lines matching a pattern.

It checks line by line and print lines matching given pattern.

Syntax:-

grep option pattern filename

Case1:

To ignore the upper and lower case while searching.

grep -i “keyword” filename

Case 2:

To print how many times given keyword present in a file.

grep -c “keyword” filename

Case 3:

To search multiple keywords in a file.

grep -e “keyword1” -e “keyword2” filename

Case 4:

To print the matching line which start with given keyword

grep “^keyword” filename

Case 5:

To print the matching line which ends with given keyword.

grep “keyword\$” filename

Case 6:

To show the line number of file with the line matched.

grep -n “keyword” filename

Case 7:

To display the lines that are not matched with the specific keyword.

grep -v "keyword" filename

Case 8:

Display all lines that contain any of the letter A,B,C,D

grep "[ABCD]" filename

egrep command in linux:-

We can use egrep command for the multiple keywords search.

egrep "keyword1|keyword2|keyword3" filename

Q2. Sed command in linux?

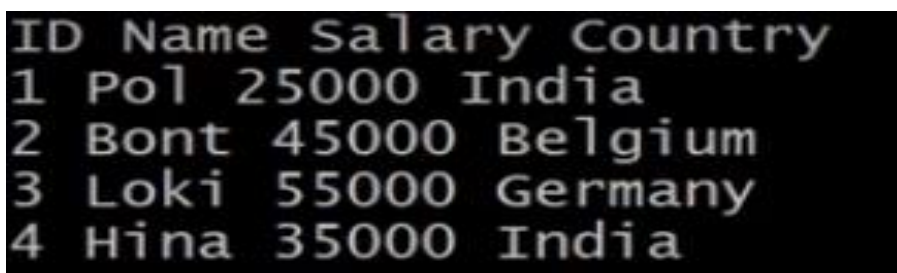
Sed stands for stream editor, which is used to search a word in the file and replace it with the word, insertion and deletion of a text.

Syntax:

sed [options] [script] [input file]

We have a file:

cat file10



```
ID Name Salary Country
1 Pol 25000 India
2 Bont 45000 Belgium
3 Loki 55000 Germany
4 Hina 35000 India
```

Basic commands:

- To globally replace a word “India” with “Bharat” in a file

sed 's/India/Bharat/g'

- To show only a given line.

sed -n '1p' file10

output:

ID Name Salary Country

- To show a last line

sed -n '\$p' file10

output:

4 Hina 35000 India

- To see all the users from India Country

sed -n '/India/p' file10

output:

1 Pol 25000 India

4 Hina 35000 India

- To delete a second line

sed '2d' file10

output:

ID Name Salary Country

2 Bont 45000 Belgium

3 Loki 55000 Germany

4 Hina 35000 India

Q3. Cut command in linux?

Cut is a command-line utility that allows you to cut line parts from specified files.

Syntax:

```
cut [option] [filename]
```

We have file:

```
cat madan.txt
```

India Delhi 10

Kolkata Punjab 40

Chennai Gujrat 30

Basic cut commands:

- To see only 2nd character

```
cut -c2 madan.txt
```

output:

**n
o
h**

- To see 1 to 4 character

```
cut -c1-4 madan.txt
```

output:

Indi

Kolk

Chen

- To print 2nd and 4th character from each line of the file

cut -c 2,4 scp.txt

output:

ni

ok

hn

Q4. Sort command in linux?

`sort` command is used to sort a file, arranging the records in a particular order.

This command does not change the actual content of the file.

Syntax:

`sort [option] [filename]`

we have a file

cat sorting.txt

Ankur Patel

Diksha mishra

Himanshu verma

Aman singh

Basic sort commands:

- To sort a file in alphabetical order

sort sorting.txt

output:

Aman singh
Ankur Patel
Diksha mishra
Himanshu verma

- To sort a file in reverse order

sort -r sorting.txt

Output:

Himanshu verma
Diksha mishra
Ankur Patel
Aman singh

- To sort a numeric content of a file we use -n

vi num.txt

345

101

544

244

:wq

sort -n num.txt

output:

101

244

345

544

- To sort a numeric content of a file in reverse order

sort -nr num.txt

output:

544

345

244

101

Q4. Paste command in linux?

paste is a command that allows you to merge lines of files horizontally.

Syntax:

```
paste [OPTION] [FILE]
```

we have a 2 file:

cat fname.txt

madan

sohan

sumit

mohit

cat lname.txt

gopal

verma

sharma

patel

Basic paste commands:

- To merge two lines in a single line of same file.

cat fname.txt | paste - -

output:

madan sohan

sumit mohit

- To merge lines of 2 files into one file.

paste fname.txt lname.txt

output:

madan gopal

sohan verma

sumit sharma

mohit patel

- To merge the first line of the first file with the first line of the second file and so on.

paste -s fname.txt lname.txt

output:

madan sohan sumit mohit

gopal verma sharma patel

- If we want two name separated by pipe “|” we use.

paste -d "|" fname.txt lname.txt

output:

madan|gopal

sohan|verma

sumit|sharma

mohit|patel

Q4. User to user Communication command in linux?

1. Write:

- It is used to send a message to another user from same terminal or different terminal.

- write utility allows a user to communicate with other user.
- When any user run the write command, the user you are writing to gets a message of the form:

message from yourname@yourhost on yourtty at hh:mm

...

- If you want to reply then run write command.
- When you terminate conversation then send EOF.
- Syntax of write:

write <user> <tty name>

where, the user parameter is mandatory and represents the username of the receiving end. The tty name specifies the terminal environment.

Example:

```
write madan
```

```
hello madan where are you
```

```
^D
```

The same message appears on the receiving end in real-time.

2. mesg

- If you don't want to receive any messages from other users, use the mesg command.

Example:

1. To allow write access to your terminal:

```
mesg y
```

2. To disallow write access to your terminal:

```
mesg n
```

3. Wall

- wall stand for write to all command in Linux system it is used to write a message simultaneously to all other logged in users.

- Only admin can execute these command.
- All users can read the message they cannot block it.
- Syntax:

wall [OPTIONS] <MESSAGE>

Example:

wall “the system will restarted in 10 minutes”

Q5. Process in Linux ?

- ❖ A process is an instance of a program.
- ❖ Each process is identified by a unique process id
- ❖ ps (process status) command is used to display all running process in linux system.
- ❖ Syntax:

ps [options]

- ❖ To see all the running process

ps -e

There are two categories of processes in Unix:

- **User processes:** They are operated in user mode.
- **Kernel processes:** They are operated in kernel mode.

Structure of process:

Q3 Explain structure of process & process control.

The structure of process is defined by data structure known as task_struct in the linux. This structure contains all necessary information to manage a process.

- **Process ID** :- A unique identifier for each process.
- **Process state** :- The current state of the process.
- **Program Counter** :- The address of next instruction to execute.
- **CPU Registers** :- The current value of CPU

register for process.

- **Memory Management Information** :- This includes pointers to the process code & stack segment.

Process Control :- Process control refers to the mechanism & commands used to manage processes running on linux.

Process Control Commands are :-

- **bg** :- Put suspended process into background. user can run a job in the background by adding "&" symbol at the end of command.

Syntax :-

bg [job]

- **fg** :- It moves a background job into the foreground.

Syntax :-

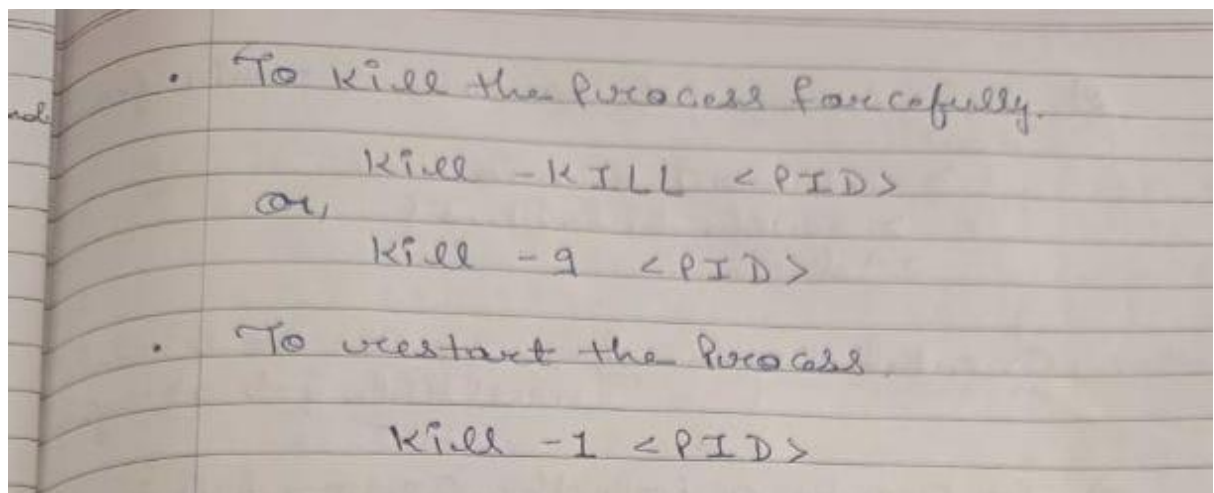
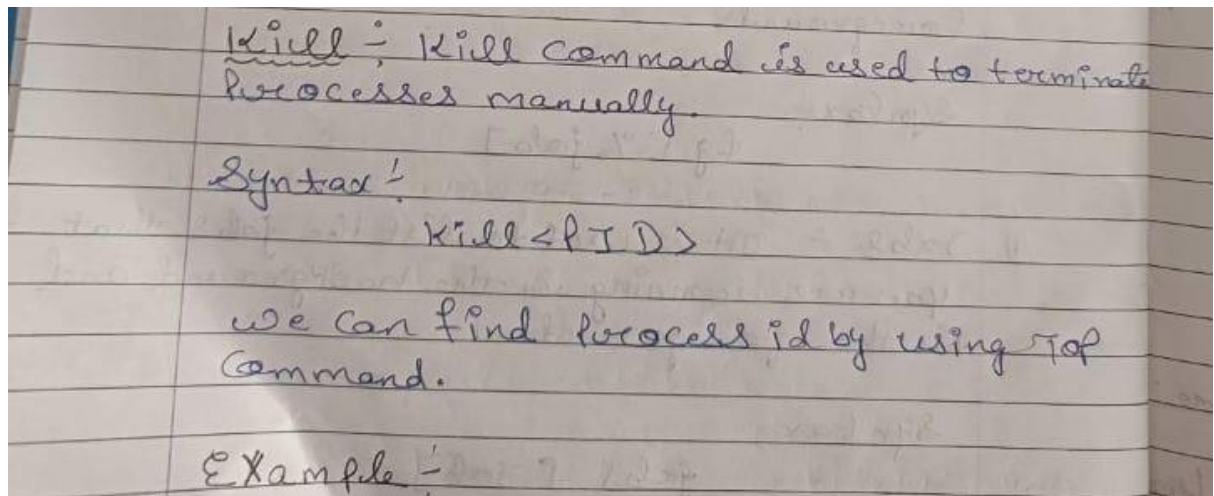
fg [%job]

- **Jobs** :- It is used to list the jobs that you are running in the background and in the foreground.

Syntax :-

jobs [Job]

Q6. kill command in linux ?



Q7. at and crontab command?

at : It schedule job once at a specified time in the future.

→ To ~~see~~ job is schedule ~~or~~ not the command is
atq.

→ To delete a job the command is :-
atrm.

Example :-

1) Schedule a job to run at 3 pm from now
at 15:00.
or,
at 3 pm.

SK

Teacher's Sign

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Roll No. _____ Page No. 10 Date. / /

> echo "abcd" > alphabet.
> ^d.

The above command create a new file name alphabet & schedule at 3 pm.

2) Schedule a job to run at 5 min from now
at now + 5 min.
> touch file1.r3
> mkdir /madan
> ^d.

The above command create a 5 empty file and create a directory name madan & schedule at 5 min from now.

3 To schedule a job to run at tomorrow 9 AM

at 9:00 AM tomorrow

>rm file100

>^d

The above command remove a file100 at 9 AM tomorrow.

4 To schedule a job to run at 30 july 10 PM

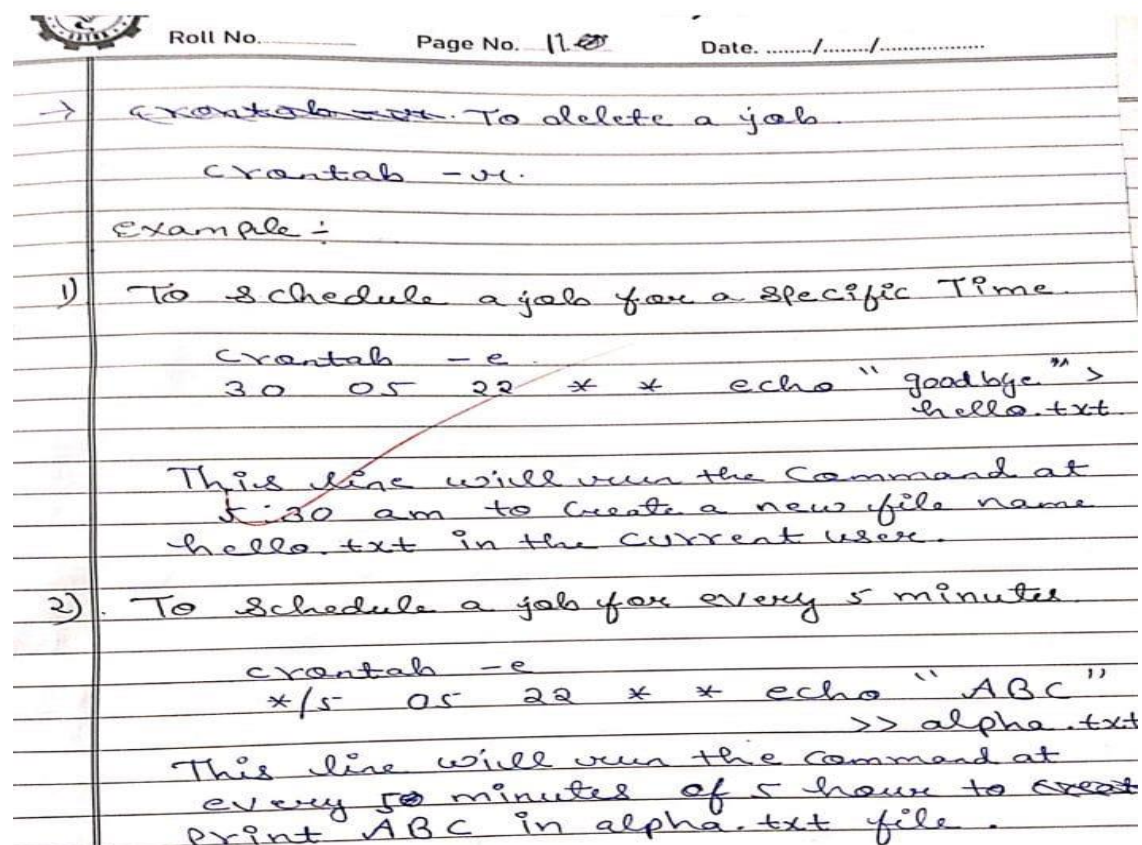
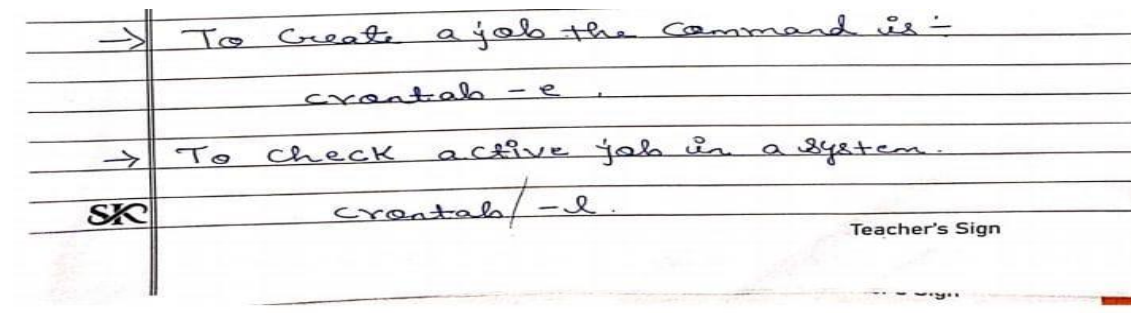
at 22:00

>sudo reboot

>^d

The above command reboot a system at 30 july
10 PM

crontab : It schedule repetitive jobs that we want to run
at regular interval.



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Roll No. _____ Page No. 12 Date. / /

3) To reboot a system at 01:2 PM.
09 13 * * * Sudo reboot.

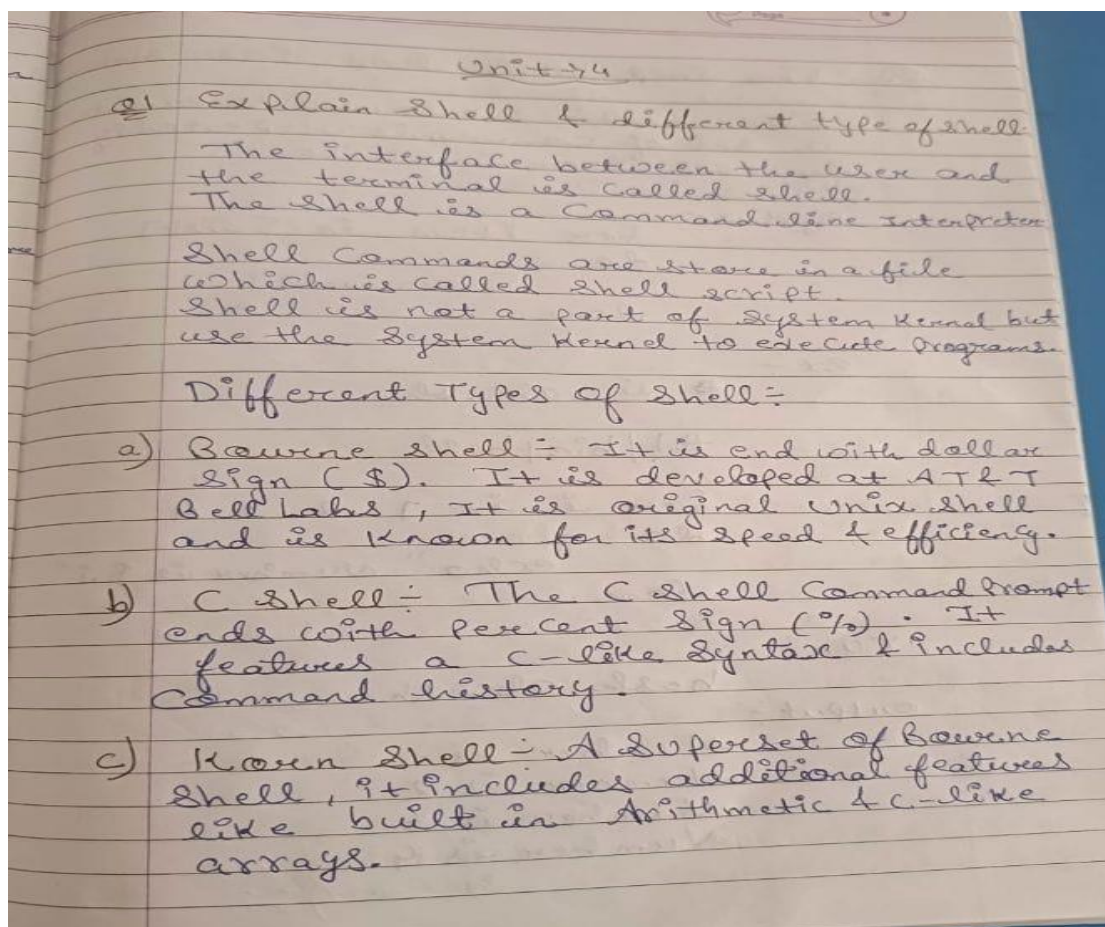
4) To Poweroff a system at 10:05 AM.
05 10 * * * Sudo poweroff.

5) To Schedule multiple Job at same time.
08 12 * * * cron file100; mkdir /dir100.

Unit-4

Q1. Explain Shell and different type of shell?

A shell provide an environment to a user to execute command and interact with kernel.



d. Z shell: It is an extended version of bourne shell with many additional features like:

- Improved variable/ Array handling
- Editing of multi line commands
- Spelling correction

Q2. Explain shell variable ?

Shell variables in linux are placeholders that store data such as filename, strings, numbers or command outputs for use in shell scripts.

Syntax:

Variable_name = value

Example:

var_name = \$(hostname)

simple Shell script :

vi abc.sh

#!/bin/bash

Name = "madan"

echo "my name is \$Name"

To run a script :

bash abc.sh

output:

my name is madan

System shell variable :

- In linux, system shell variables are predefined variables that store information about the system environment, user session and so on.
- These variable are set by the system and can be accessed and used by user in a scripts.

Some system variable are:

1. HOME
2. PATH
3. HOSTNAME
4. PWD
5. UID

Simple shell script:

vi System.sh

```
#!/bin/bash
```

```
HOSTNAME = $(hostname)
```

```
echo "name of this machine is $HOSTNAME"
```

To run a script :

```
bash System.sh
```

output:

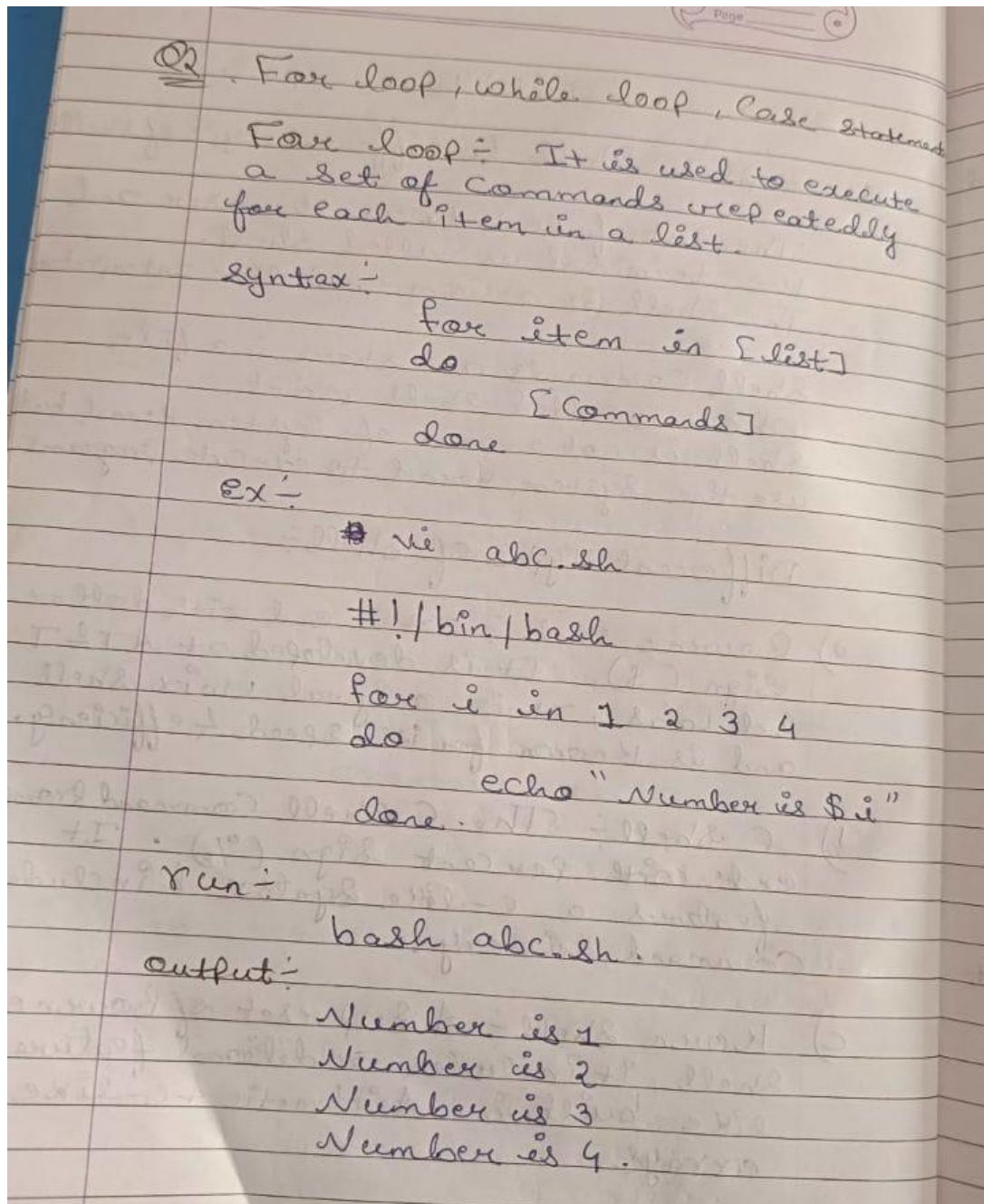
```
name of this machine is machine10
```


Q3. Looping statement in linux?

There are 3 looping statement in shell programming:

1. for loop
2. while loop
3. until loop

1. for loop:



2. while loop

- while loop allow code to be executed repeatedly based on given boolean condition.
- The loop terminates when the condition become false.

Syntax:-

```
while [Condition]
do
    [Commands]
done
```

Ex:-

```
vi madan.sh
#!/bin/bash

Count=0
num=3

while [[ $Count -le $num ]]
do
    echo "value of Count
    Variable is $Count"
    let Count++
done
```

run:-

```
bash madan.sh
```

output:-

```
Value of Count Variable is 0
Value of Count Variable is 1
Value of Count Variable is 2
Value of Count Variable is 3
```

3. until loop

- The until loop execute many times as the condition evaluates to false.
- The loop terminate when the condition become true.

Syntax:

```
until [condition]
do
    [commands]
done
```

ex:-

```
vi abc.sh
```

```
#!/bin/bash
```

```
count = 1
```

```
until [ $count -gt 4 ]
```

```
do
```

```
    echo "$count"
```

```
    let count++
```

```
done
```

run the script:

bash abc.sh

output:

0
1
2
3
4

Q4. Conditional statement in linux ?

1. if statement

This will process if specified condition is true.

Syntax:

```
if [ expression ]  
then  
    statement  
fi
```

Example:

```
vi sam.sh  
#!/bin/bash  
if [ $(whoami) = 'root' ]  
then  
    echo "you are root"  
fi
```

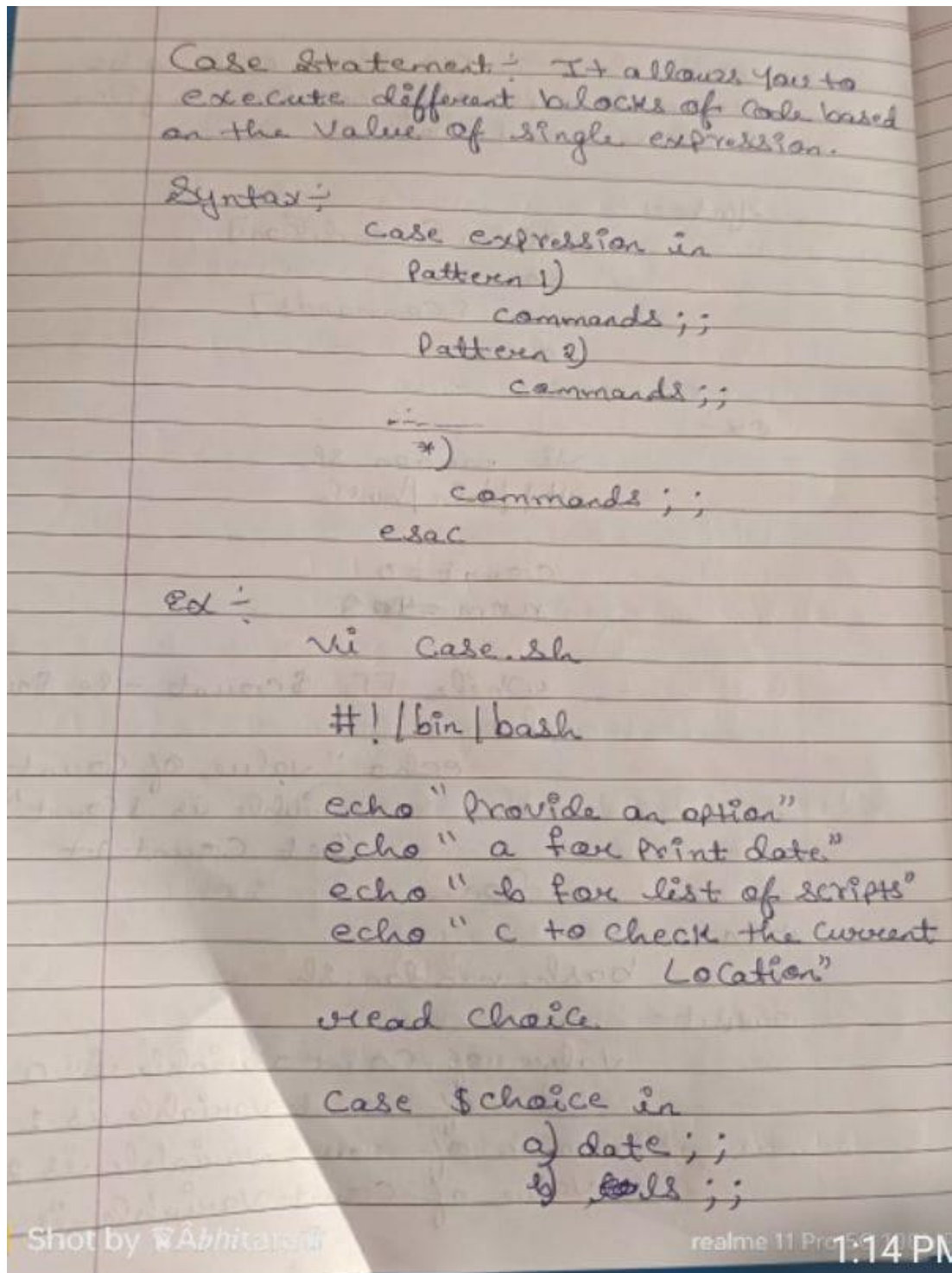
run the script:

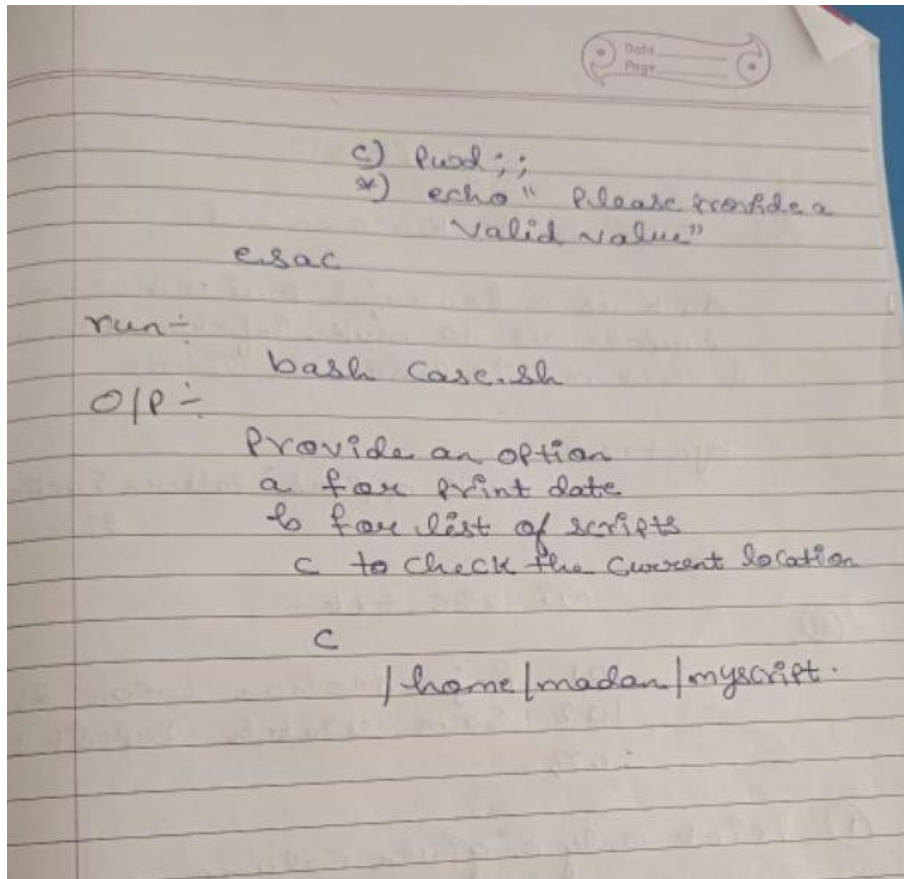
bash sam.sh

output:

you are root

2. case statement





Unit-5

Q1. Explain AWK command ?

AWK stands for Aho, Weinberger and Kernighan.

AWK Command is a powerful tool whose main purpose is to make information retrieval and text manipulation in linux.

Syntax:

awk options 'pattern {action}' file_name

Terms used:

NR = No. of row / record

NF = No. of fields / column

\$0 = Print everything

\$1, \$2 = Field no.

vi abc.txt

101 Raju Mahan Loan 2500

102 Sam Watch Deposit 5000

103 Alex Rohan Manager 4000

104 Paul Sunil Loan 6500

!wq

Examples

1) Print only a given column.

SK

awk '{print \$3}' abc.txt

Teacher's Sign

output:

Mahan
Watch
Rohan
Sunil

2) Print last column.

awk '{print \$NF}' abc.txt

output:

2500
5000
4000
6500

3) To search a word

awk '/Loan/' '{print \$0}' abc.txt

output:

101 Raju Mahan Loan 2500
104 Paul Sunil Loan 6500

4) Print only a given line no.

awk 'NR == 2 {print NR, \$0}' abc.txt

output:

2 102 Sam Watch Deposit 5000

SK

5) Print range of lines

```
awk 'NR==3, NR==4 {print NR, $0}' abc.txt
```

output.

```
103 Alex Rohan Manager 4000
104 Paul Sunil Loan 6500
```

6) Search Multiple words.

```
awk '/sam | Paul {print $0}' abc.txt
```

output.

```
102 Sam Watch Deposit 5000
104 Paul Sunil Loan 6500
```

7) Check a given char is present in column.

```
awk ' $2 ~ /a/ {print $0}' abc.txt
```

output.

```
101 Raju Mahan Loan 2500
102 Sam Watch Deposit 5000
104 Paul Sunil Loan 6500
```

8) Ignore Case while Searching.

```
awk 'BEGIN {IGNORECASE=1} /deposit/ {print $0}' abc.txt
```

output.

```
102 Sam Watch Deposit 5000
```

Teacher's Sign

5:43 PM